

SIMATS ENGINEERING
ASSESSMENT TOOL 1: Short Answer Test

COURSE NAME: Cloud Computing and Big Data Analytics **COURSE CODE:** CSA15

COURSE OUTCOME COVERED

CO1: *Apply cloud computing technologies including hardware-software evolution, virtualization, web services, and service models to develop cloud-based solutions. (BL2)*

Assessment Tool Weightage: 40%

Dave's Taxonomy: Imitation (Level 1)
Question Count: 5

Total Marks: 25

Code of Conduct

I K Charan / 192524488 (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.

I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student: K Charan

Assessment Tasks

Short Answer Test

1. Define cloud computing and explain any four characteristics that make it different from traditional on-premise computing.
2. Trace the evolution of cloud computing from hardware evolution to internet software evolution, and show how these changes enabled modern cloud platforms.
3. Explain the role of server virtualization in cloud computing. Differentiate between virtualization as an enabling technology and cloud computing as a service model.
4. Discuss the role of **data centers** in delivering cloud services..
5. Differentiate IaaS, PaaS, SaaS, and XaaS with one suitable real-world example for each model.

Short Answer Test Rubric

Criteria	Weightage	Level 4 - Exemplary	Level 3 - Proficient	Level 2 - Developing	Level 1 - Beginning
Conceptual Accuracy	30%	Accurate definitions and precise explanation of all key terms	Mostly accurate with minor gaps	Partial understanding with noticeable errors	Incorrect or irrelevant explanation
Coverage of Concepts	20%	Covers all expected sub-points completely	Covers most expected points	Covers only basic points	Very limited coverage
Use of Examples	15%	Uses highly relevant cloud examples to strengthen the answer	Uses relevant examples with minor mismatch	Uses vague or partially relevant examples	No useful examples
Comparison / Differentiation	20%	Clearly distinguishes concepts with strong logic	Reasonable differentiation with minor overlap	Comparison is weak or incomplete	Fails to differentiate concepts
Clarity of Expression	15%	Clear, organized, and concise answer writing	Generally clear with few issues	Understandable but poorly organized	Difficult to follow

CO-1

Assessment Tool-3

Name :- K. Charan

Reg No :- 192524488

Course Name :- Cloud Computing

Course Code :- CSA-1507

Date :- 28-07-2026

1A:- Definition :-

Cloud Computing is the delivery of computing services such as servers, storage databases, networking, software, and analytics over the internet. It allows users to access resources on demand without owning physical infrastructure.

Characteristics :-

1. On - Demand Self-Service :- Users can create and use computing resources whenever required without human intervention.
2. Broad Network Access :- Cloud services are accessible through the internet using devices such as laptop, smart-phones, and tablets.
3. Resource pooling :- The cloud provides shared computing resources among multiple users using virtualization, improving efficiency.

H. Rapid Elasticity:-

Resources can be increased or decreased automatically according to workload, making the cloud highly Scalable.

Conclusion:-

These features make cloud computing flexible, cost-effective, and easier to manage than traditional on-premise system.

2A:-

Cloud computing evolved Through Several stages:-

1. Mainframe computers (1960s):-

Many users shared on the a single powerful computer through terminals

2. Personal computers (1980s):-

Individual users started using their own computers

3. Client - Server Architecture:-

Applications were divided between client systems and central servers.

H. Internet Revolution (1990s):-

The internet enable communication and online services worldwide.

5. Virtualization Technology:-

One physical server could run multiple virtual machines, improving resource utilization.

6. cloud computing (2000s onwards) :-

Companies like AWS, Microsoft Azure, and Google cloud began offering computing resources as online services.

Result:-

Hardware improvement, virtualization, and internet technologies together enabled today's scalable and reliable cloud platforms.

3A1

Role of Server Virtualization:-

Server virtualization divides one physical server into multiple virtual machine (VMs) Each VM works independently with its own operating system and applications.

Benefits:-

- Better hardware utilization
- Lower infrastructure cost
- Easier backup and recovery
- Faster deployment of applications
- Support cloud scalability

Difference:-

Aspect	Virtualization	cloud computing
Def	Technology to wh OS on the hardware	Service model to delivers relays over internet
Scope	Can be used on-prem	Always delivered as a services.
features	only resource abstract	Add self Service billing elasticity.

Conclusion:-

Virtualization is the foundation, while cloud computing is the service built on top of it.

HA- Data centres are the physical infrastructure behind cloud

1. Resource Hosting:-

Contain servers, storage, and networking equipment used to run cloud services.

2. Reliability:-

provide power backup, cooling and redundancy to ensure 99.99% uptime.

3. Global presence:-

Cloud providers have regions and availability zones for low latency and disaster recovery.

4. Security:-

Physical security, firewall, and compliance certifications are managed by data center.

5. Connectivity:-

Highspeed internet and interconnection allows users to access services.

Conclusion:-

Data centres are the backbone of cloud computing because they provide reliable, secure and uninterrupted cloud services.

SA	<u>Service model</u>	<u>Description</u>	<u>Example</u>
	1. IaaS (Infrastructure as a service).	Provides virtual server storage and networking users manage the operating system and application.	Amazon EC2

Service model	Description	Example.
2. PaaS (Platform as a Service)	Provides a platform for developing and deploying applications without managing	Google App Engine.
3. SaaS (Software as a Service)	Provides a platform for developing software through a web browser.	Google Docs (or) Microsoft 365
4. XaaS (Anything as a Service)	Refers to any IT service delivered over the cloud	Microsoft Azure Services.

Conclusion:-

These services models provide different levels of control and management based on user needs.